

BudgetApp - User Guide

Personal Budget Management Software

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1. Setup and Installation

Prerequisites

- Node.js (v18 or higher)
- PostgreSQL (v15 or higher)
- npm (comes with Node.js)

Steps

- 1 Create a PostgreSQL database named `budgetapp`

2 Open the `.env` file and update the database connection string:

```
DATABASE_URL=postgresql://username:password@localhost:5432/budgetapp
```

3 Open a terminal in the project folder and run:

```
npm install
```

4 Start the server:

```
npm start
```

5 Open your browser and go to `http://localhost:3000`

The database tables are created automatically when the server starts for the first time. Default categories (Salary, Freelance, Food, Transport, Shopping, Bills, Entertainment, Health, Other) are added when you register.

2. Creating an Account

1 Open the app in your browser. You will see the login page.

2 Click "**Don't have an account? Sign up**" at the bottom.

3 Enter your name, email, and a password.

4 Click "**Sign Up**". You will be redirected to the Dashboard automatically.

3. Logging In

1 Enter your email and password on the login page.

2 Click "**Login**".

3 If the credentials are correct, you will be taken to the Dashboard.

If you enter a wrong email or password, a red error message will appear below the form.

4. Dashboard

The Dashboard is the main screen you see after logging in. It shows:

Card	Description
Balance	Your total income minus total expenses
Income	Total money you earned
Expenses	Total money you spent
Active Budgets	Number of budgets currently being tracked
Active Goals	Number of savings goals in progress

Below the cards, you can see your **5 most recent transactions**.

5. Managing Transactions

Adding a Transaction

1 Click "**Transactions**" in the navigation bar.

2 Click the "**+ Add Transaction**" button at the top.

3 Fill in the form:

- **Type:** Choose "Income" or "Expense"
- **Category:** Select a category (changes based on type)
- **Amount:** Enter the amount in EGP
- **Date:** Pick the date
- **Description:** Optional note about the transaction

4 Click "**Save**".

Editing a Transaction

1 Find the transaction in the table.

2 Click the "**Edit**" button next to it.

3 The form will appear pre-filled with the transaction data. Make your changes and click "**Save**".

Deleting a Transaction

1 Click the "**Delete**" button next to the transaction.

2 Confirm the deletion in the popup.

6. Creating and Tracking Budgets

Creating a Budget

1 Click "**Budgets**" in the navigation bar.

2 Click "**+ Create Budget**".

3 Fill in the form:

- **Category:** Which expense category to track
- **Amount:** Your monthly spending limit in EGP
- **Month/Year:** Which month this budget applies to
- **Alert %:** At what percentage you want to be warned (default 80%)

4 Click "**Save**".

Budget Progress

Each budget shows a progress bar:

- **Green** = spending is under control
- **Orange** = nearing your alert threshold
- **Red** = you exceeded the budget

7. Budget Alerts

When you add an expense, the system automatically checks if you are close to or over your budget for that category:

Status	What Happens
Under threshold	Nothing happens, progress bar stays green
Near limit (e.g. 80%+)	Orange warning notification is created
Exceeded	Red alert notification is created

You can see alerts in the Notifications page. The badge on "Notifications" in the navbar shows how many unread alerts you have.

8. Savings Goals

Creating a Goal

- 1 Click **"Goals"** in the navigation bar.
- 2 Click **" + New Goal "**.
- 3 Enter a name, target amount, and optional deadline.
- 4 Click **"Save"**. The goal appears with a 0% progress bar.

Adding Money to a Goal

1 Find the goal card.

2 Enter an amount in the small input field.

3 Click "**Add Money**".

4 The progress bar updates. When you reach the target, a congratulations message appears and the goal is marked as completed.

9. Reports and Analytics

1 Click "**Reports**" in the navigation bar.

2 Optionally select a date range (From and To).

3 Click "**Generate**".

The report shows:

- **Total Income / Total Expenses / Savings Rate** - summary cards at the top
- **Spending by Category** - table showing how much you spent per category
- **Monthly Trend** - income and expenses broken down by month
- **Top expense category** - which category you spend the most on

10. Notifications

1 Click "**Notifications**" in the navigation bar (shows a red badge with unread count).

2 You will see a list of all budget alerts with timestamps.

3 Click "**Mark all as read**" to clear the badge counter.

11. Technology Stack

Layer	Technology
Frontend	Alpine.js, Bootstrap 5, HTML5
Backend	Node.js, Express.js
Database	PostgreSQL (via pg library)
Authentication	JWT (JSON Web Tokens), bcryptjs

Project Structure

File/Folder	Purpose
server/index.js	Main server entry point
server/routes/	API route handlers (auth, transactions, budgets, goals, reports, etc.)
server/models/	Database connection and schema
server/middleware/	JWT authentication middleware
public/index.html	Frontend single page application
public/js/app.js	Alpine.js application logic
public/css/app.css	Custom styles
.env	Environment variables (DB connection, JWT secret)

12. Application Screenshots

Login Page

Users enter their email and password to access the app.

Personal Budget Management

Email

Password

[Login](#)

Dashboard

The main screen shows balance, income, expenses, active budgets/goals, and recent transactions.

BudgetApp

DashboardTransactionsBudgetsGoalsReports

Notifications3Logout

Dashboard

Balance

EGP 13250.00

Income

EGP 17000.00

Expenses

EGP 3750.00

Active Budgets

3

Active Goals

3

Recent Transactions

Date	Category	Type	Amount	Description
2026-04-19T22:00:00.000Z	Freelance	income	+1500.00	Website design gig
2026-04-14T22:00:00.000Z	Shopping	expense	-350.00	Amazon order
2026-04-11T22:00:00.000Z	Food	expense	-400.00	Restaurant dinner
2026-04-09T22:00:00.000Z	Freelance	income	+2500.00	Freelance project
2026-04-09T22:00:00.000Z	Food	expense	-900.00	Food shopping

Transactions

All income and expense records with edit and delete options.

BudgetApp

DashboardTransactionsBudgetsGoalsReports

Notifications3Logout

Transactions

+ Add Transaction

Date	Category	Type	Amount	Description	Actions
2026-04-19T22:00:00.000Z	Freelance	income	1500.00	Website design gig	EditDelete
2026-04-14T22:00:00.000Z	Shopping	expense	350.00	Amazon order	EditDelete
2026-04-11T22:00:00.000Z	Food	expense	400.00	Restaurant dinner	EditDelete
2026-04-09T22:00:00.000Z	Freelance	income	2500.00	Freelance project	EditDelete
2026-04-09T22:00:00.000Z	Food	expense	900.00	Food shopping	EditDelete
2026-04-08T22:00:00.000Z	Health	expense	300.00	Doctor visit	EditDelete
2026-04-07T22:00:00.000Z	Entertainment	expense	250.00	Netflix and cinema	EditDelete
2026-04-06T22:00:00.000Z	Bills	expense	400.00	Electricity and internet	EditDelete
2026-04-04T22:00:00.000Z	Shopping	expense	600.00	New shoes	EditDelete
2026-04-02T22:00:00.000Z	Transport	expense	200.00	Uber rides	EditDelete
2026-04-01T22:00:00.000Z	Food	expense	350.00	Grocery shopping	EditDelete
2026-03-31T22:00:00.000Z	Salary	income	8000.00	Monthly salary	EditDelete
2026-03-31T22:00:00.000Z	Salary	income	5000.00	Salary	EditDelete

Add Transaction Form

Users select type, category, amount, date, and description. Budget is checked automatically on expenses.

BudgetApp

DashboardTransactionsBudgetsGoalsReports

Notifications3Logout

Transactions

+ Add Transaction

New Transaction

Type

Expense

Category

Bills

Amount

150

Date

05/05/2026

Description

Coffee and snacks

SaveCancel

Date	Category	Type	Amount	Description	Actions
2026-04-19T22:00:00.000Z	Freelance	income	1500.00	Website design gig	EditDelete
2026-04-14T22:00:00.000Z	Shopping	expense	350.00	Amazon order	EditDelete
2026-04-11T22:00:00.000Z	Food	expense	400.00	Restaurant dinner	EditDelete
2026-04-09T22:00:00.000Z	Freelance	income	2500.00	Freelance project	EditDelete
2026-04-09T22:00:00.000Z	Food	expense	900.00	Food shopping	EditDelete
2026-04-08T22:00:00.000Z	Health	expense	300.00	Doctor visit	EditDelete
2026-04-07T22:00:00.000Z	Entertainment	expense	250.00	Netflix and cinema	EditDelete
2026-04-06T22:00:00.000Z	Bills	expense	400.00	Electricity and internet	EditDelete
2026-04-04T22:00:00.000Z	Shopping	expense	600.00	New shoes	EditDelete
2026-04-02T22:00:00.000Z	Transport	expense	200.00	Uber rides	EditDelete
2026-04-01T22:00:00.000Z	Food	expense	350.00	Grocery shopping	EditDelete

Budgets

Progress bars change color: green (safe), orange (nearing limit), red (exceeded).

BudgetApp

DashboardTransactionsBudgetsGoalsReports

Notifications3Logout

Budgets

+ Create Budget

Food

4/2026

Spent: EGP 1650.00Limit: EGP 1000.00

165%

exceeded

Delete

Shopping

4/2026

Spent: EGP 950.00Limit: EGP 500.00

190%

active

Delete

Bills

4/2026

Spent: EGP 400.00Limit: EGP 600.00

67%

active

Delete

Entertainment

4/2026

Spent: EGP 250.00Limit: EGP 400.00

63%

active

Delete

Savings Goals

Goals show progress. When target is reached, a congratulations message appears.

BudgetApp

DashboardTransactionsBudgetsGoalsReports

Notifications3Logout

Goals

+ New Goal

Vacation Trip

Deadline: 2026-08-14T21:00:00.000Z

Saved: EGP 0.00Target: EGP 15000.00

Amount

Add Money

Delete

New Laptop

Deadline: 2026-08-31T21:00:00.000Z

Saved: EGP 15000.00Target: EGP 35000.00

43%

Amount

Add Money

Delete

Emergency Fund

Deadline: 2026-12-30T22:00:00.000Z

Saved: EGP 12000.00Target: EGP 20000.00

60%

Amount

Add Money

Delete

Emergency Fund

Deadline: 2026-12-30T22:00:00.000Z

Saved: EGP 10000.00Target: EGP 10000.00

100%

Congratulations! Goal completed!

Reports

Income vs expenses by category with date range filter, savings rate, and monthly trends.

BudgetApp

DashboardTransactionsBudgetsGoalsReports

Notifications3Logout

Reports

From

04/01/2026

To

04/30/2026

Generate

Total Income
EGP 17000.00

Total Expenses
EGP 3750.00

Savings Rate
77.9%

Spending by Category

Category	Type	Total (EGP)
Salary	income	13000.00
Freelance	income	4000.00
Food	expense	1650.00
Shopping	expense	950.00
Bills	expense	400.00
Health	expense	300.00
Entertainment	expense	250.00
Transport	expense	200.00

Monthly Trend

Month	Income	Expense
2026-04	-	3750.00
2026-04	17000.00	-

Notifications

Budget alerts appear automatically when spending nears or exceeds limits.

Notifications

Mark all as read

- Alert

You have exceeded your budget for this category! Spent 1650.00 out of 1000.00

5/5/2026, 6:00:44 AM
- Alert

You have exceeded your budget for this category! Spent 1250.00 out of 1000.00

5/5/2026, 6:00:43 AM
- Warning

You are nearing your budget limit. Spent 900.00 out of 1000.00 (90%)

5/4/2026, 10:47:56 PM